



IILM UNIVERSITY

School of Computer Science & Engineering

✦ AICTE & ICAC Approved Internship Program

INTERNSHIP REPORT PRESENTATION

Web Development Internship — SMS Spam Detection

SUBMITTED BY

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Roll No.: 25SCS1003001466

Programme: B.Tech – Computer Science & Engineering

Institution: IILM University | Batch 2025–2029



ORGANIZATION

Codec Technologies Pvt. Ltd.

DURATION

01 July 2026 – 30 July 2026

Codec Technologies Internship

The internship at **Codec Technologies Pvt. Ltd.** provided comprehensive practical exposure to modern software development, web workflows, and data-driven machine learning models.

Key Areas of Learning

- ✓ Python programming & notebook-based development
- ✓ Data handling, cleaning, and exploratory analysis
- ✓ Natural Language Processing (NLP) & text processing
- ✓ Machine Learning algorithms & text classification
- ✓ Model training, soft voting evaluation & persistence

Program Overview

- ✓ **Role:** Web Developer Intern
- ✓ **Duration:** 1 Month (01 July 2026 – 30 July 2026)
- ✓ **Mode:** Pan India – Hybrid Workflow
- ✓ **Approval:** AICTE & ICAC Approved Program
- ✓ Successfully bridged academic theory with practical application

Problem Statement



Bridging Development Skills with Machine Learning

Unwanted and fraudulent SMS messages create serious security, privacy, and communication challenges. Manual identification of spam messages is highly inefficient when processing large volumes of real-time communication data.

HAM Message

Legitimate and safe communication message received from trusted sources.

SPAM Message

Unwanted, promotional, or fraudulent message requiring automated filter blocks.

Development Workflow

Dataset



Data Cleaning



Text Preprocessing



TF-IDF Extraction



Model Training



Spam/Ham Result

Project Objectives

Key Objectives

- ✓ Understand and inspect a real-world SMS dataset
- ✓ Clean raw text data by removing noise & duplicates
- ✓ Perform Exploratory Data Analysis (EDA)
- ✓ Apply NLP techniques (Tokenization, Stopwords, Stemming)
- ✓ Convert text into numerical features using TF-IDF

Model Development & Goal

- ✓ Train and compare multiple Machine Learning algorithms
- ✓ Enhance performance using Soft Voting Ensemble learning
- ✓ Evaluate system using Precision and Accuracy metrics
- ✓ Save trained models with Pickle for seamless reuse
- ✓ **Final Goal:** Build a highly accurate automated SMS spam classifier

Technologies & Tools

Programming & IDE

Python Jupyter Notebook Google Colab
Git & GitHub

Data Handling

Pandas NumPy DataFrames

NLP Tools

NLTK Regular Expressions Tokenization
Stopwords PorterStemmer

Machine Learning

Scikit-learn Naive Bayes SVC
Extra Trees Random Forest
Logistic Regression

Feature Extraction

CountVectorizer TF-IDF Vectorizer
Feature Scaling

Model Persistence

Pickle Model Serialization
Pipeline Deployment

System Architecture

1. DATA SOURCE

SMS Dataset (5,572 raw records) → Data Inspection & Structure Analysis

2. DATA CLEANING

Remove null/unnecessary columns • Rename useful fields • Remove duplicate entries

3. PREPROCESSING

Lowercase conversion • Tokenization • Punctuation/Noise removal • Stemming via NLTK

4. FEATURE VECTOR

TF-IDF Vectorization → Numerical Feature Matrix → 80:20 Train-Test Split

5. ML ENSEMBLE

Soft Voting Classifier (Support Vector Classifier + Multinomial NB + Extra Trees)

6. CLASSIFICATION

Final Output: Automated HAM (Legitimate) vs. SPAM Classification

Internship Learning Journey

PHASE 1

Programming

- Python fundamentals
- Notebook environment
- Development setup
- Debugging code

PHASE 2

Data Prep

- Dataset inspection
- Removing duplicates
- Handling missing values
- Exploratory analysis

PHASE 3

NLP Pipeline

- Text normalization
- Tokenization
- Stopword filtering
- TF-IDF extraction

PHASE 4

ML Training

- Train-Test splitting
- Comparing classifiers
- Evaluation metrics
- Accuracy analysis

PHASE 5

Optimization

- Ensemble learning
- Soft voting implementation
- Pickle persistence
- Documentation

Key Outcomes & Data Insights

Internship Growth

- ✓ Strengthened core Python & Web development workflows
- ✓ Built systematic debugging and problem-solving habits
- ✓ Gained experience in technical documentation & model deployment

Project Outcomes

- ✓ Cleaned and preprocessed unstructured text dataset
- ✓ Integrated TF-IDF feature extraction with multi-model evaluation
- ✓ Applied Soft Voting ensemble for maximum performance

Dataset Statistics Breakdown

Metric Metric Category	Original Count	Cleaned Count	Ham (Legitimate)	Spam (Fraudulent)
SMS Messages Count	5,572	5,169	4,516 (87.4%)	653 (12.6%)



Model Performance & Conclusion

Soft Voting Classifier Ensemble

The optimal performance was achieved by combining three distinct algorithms:

- SVC
- Multinomial Naive Bayes
- Extra Trees

Delivered superior classification accuracy over individual baseline models.

98.16%

Model Accuracy

99.17%

Model Precision



Internship Completion Summary

Successfully completed the **1-Month AICTE & ICAC Approved Web Development Internship** at **Codec Technologies Pvt. Ltd.** (01 July 2026 – 30 July 2026), satisfying all project milestones and institutional requirements.



Certificate of Completion —issued 30 July 2026



Internship Offer Letter

Thank You

LEKISHA

B.TechCSE | IILMUniversity, GreaterNoida

Internship Report Presentation — Web Development Internship Codec Technologies Pvt. Ltd.